Precautions for DIY Robotic Arm

Note: When the robotic arm grips objects, you need to control the gripper angle. Unreasonable angle settings may cause servo stall and burnout.

Here is a compiled gripper angle correspondence table, recording the approximate servo angles needed for gripping objects of 0.5cm size increments.

You can adjust the angle settings during gripping based on this table to avoid servo stall situations.

Object Length (Unit: cm)	Servo Angle (Unit: degrees)
0	180
0.5	176
1.0	168
1.5	160
2.0	152
2.5	143
3.0	134
3.5	125
4.0	115
4.5	105
5.0	95
5.5	80
6.0	57
6.0-6.4	0-57

For example, the visual recognition blocks we provide have length, width, and height of 3cm, so setting the servo angle to 134° will allow gripping. Do not set excessive angles.

3CM